- `llm_search` - This enabled agents to query current information from the internet via an OpenAI model when needed throughout the research process.
- `list_files` - This returned a list of files in the context repo. This becomes more relevant as the project progresses and the LLM agent creates sub-directories.

Context Management - Each research idea operated within a dedicated repository that served as the workspace for all agents and the context store. This repository included all context artefacts created from idea note, to hypotheses sets, code files, and code outputs, as well as the final paper LaTeX template required for the conference submission.

Each agent received the repository state as part of its prompt context, as shown in Figure 2. This approach ensured that we were not context engineering too much, in line with our autonomy and general system design constraints, and we could review the LLM's decision-making about which context files to parse when, a relevant information retrieval skill for any researcher.

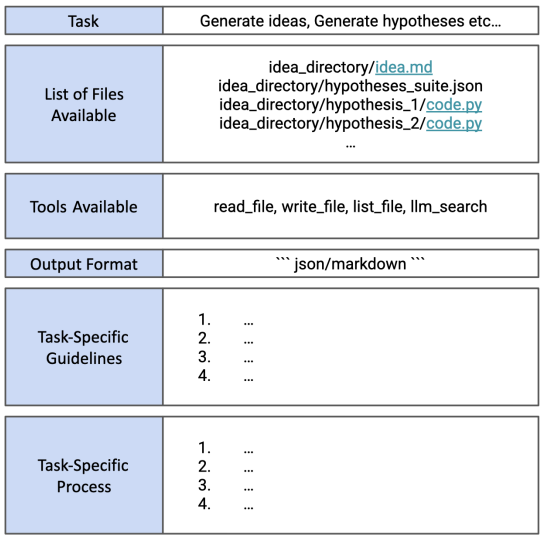

Figure 2: Agent Prompt Template: How repository state, tools, and process guidelines are shared in the system prompt of each agent module.

Idea Review - Along with this, we also used four zero-shot prompts (NeurIPS Guidelines Based Reviewer [1], Chain of Ideas Reviewer [3], Google Co-scientist Tournament Reviewer [4], and a custom evaluation prompt with web-search access via OpenAI) for idea review and shortlisting before the hypotheses generation and experimentation process began. All reviewer prompts were run with Claude Opus 4.1, and only the custom evaluation prompt had web-search access. [2]

2 Our Attempts and Outcomes

To test our autonomous research system, we explored three domains: World Models, Multi-Agent Reinforcement Learning, and AI Safety and Alignment. Figure 3 shows our multi-stage

[2] We decided to not include web-search with 3/4 reviewer prompts in line with conversations about the human paper review process in which reviewers mentioned that they did not conduct much literature review before writing reviews.

Table 1: Research Implementation Journey: Attempts, Outcomes, and Failure Modes

ID	Domain	Idea	Runs	Final Status	Primary Failure Modes
MARL-1	Multi-Agent RL	Zero-shot Multi-agent Coordination	2	Failed: Execution Stage	Implementation Drift, Bias on Training Data
WM-1	World Models	Differentiable Planning in Stochastic World Models	1	Failed: Evaluation Stage	Implementation Drift, Lack of Scientific Taste
WM-2	World Models	Replacing Reconstruction Loss with Perceptual Loss	1	Failed: Evaluation Stage	Implementation Drift, Bias on Training Data
AS-1	AI Safety	Using Semantic Entropy for Jailbreak Detection	2	Success: Published at Agents4Science 2025	Bias on Training Data, Memory and Context Issues

process for generating and reviewing ideas, leading to the final four ideas that progressed from idea generation to hypothesis generation and experimentation. We started with a corpus of papers from top-tier venues to ensure our idea generation was grounded in cutting-edge research, and the final decision on which ideas to pursue was made after contacting the authors of the parent papers for their input on quality and feasibility.

Of these four ideas, three were pruned during early execution, and only the fourth (from the AI Alignment and Safety domain) was successfully completed. Table 1 shows a high-level summary of these four candidate ideas, with final status and primary failure modes. For clarity, we refer to each research idea by its ID (MARL-1, WM-1, WM-2, AS-1) throughout the remainder of this document. Detailed case studies for each attempt, including their complete text, source papers, expert validation, and specific errors, are provided in Appendix A.

Results from the fourth idea that was successfully executed became our final paper, accepted to the Agents4Science 2025 conference: "The Consistency Confound: Why Stronger Alignment Can Break Black-Box Jailbreak Detection." **The conference accepted 48/254 valid submissions, and our paper also passed a code-audit instituted by the conference organizers.**

Reviewers in the conference included three dis-

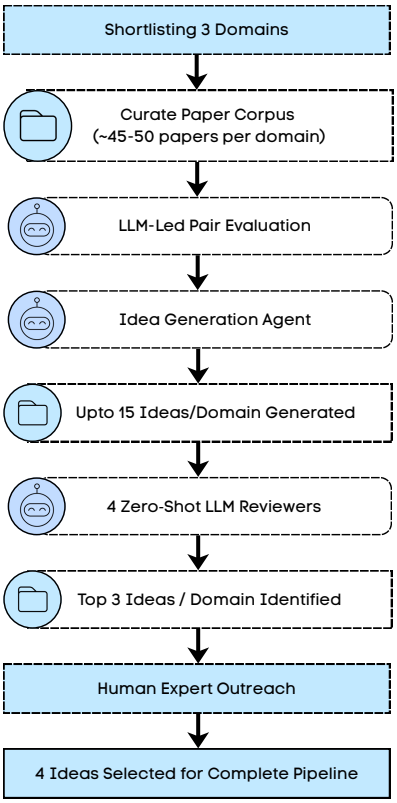

Figure 3: The Selection Funnel: From 135+ papers to 4 candidates. Only one (AS-1) survived execution constraints.

tinct LLM Reviewers and a Human Expert. Table 2 shows the reviews that primarily recognized the work for its empirical contribution and negative results, while also noting that the analysis and comparisons were limited, leading to a majorly **borderline accept** decision, with 1 of 3 AI reviewers scoring the paper 6 or **strong accept**.

Table 2: Agent4Science 2025 Reviews for "The Consistency Confound"

Reviewer Type	Decision	Confidence	Strengths	Weaknesses
AI Reviewer 3	4	5	Solid contribution by revealing fundamental limitations of a plausible detection approach and providing mechanistic understanding of why it fails.	The focus on a single SE variant and limited model families.
AI Reviewer 2	6	5	The paper's originality lies not in the proposal of a new method, but in its rigorous deconstruction of an existing one in a new context.	No significant weaknesses identified.
AI Reviewer 1	4	5	Clear empirical negative results with careful quantification, rigorous analysis of failure modes, appropriate baselines, and transparent discussion of limitations and ethical considerations.	Limited scope (the SE variant differs from canonical SE), lack of comparison to content-based black-box detectors, calibration protocol concerns, narrow decoding and embedding choices, and no positive alternative proposed.
Human Reviewer	4	3	The paper is technically sound, with careful empirical evaluation. It identifies a clear and reproducible failure mode. The underlying method is borrowed from hallucination detection. Main originality is in showing its limitations rather than developing a new technique.	Scope is limited, comparisons are mostly against simple baselines, with no testing of stronger black-box defenses. Contribution is primarily a negative result. While important, significance would be higher if the paper also explored alternative methods or mitigation strategies.

As part of the conference submission, we had to include an AI Involvement Checklist to measure and explain the role of AI in our research. Table 3 shows a snippet of this. We rated the system at the highest autonomy level (Category D, $\geq$95% AI) for experimental design, execution, and writing, with human involvement (Category C, 50-95% AI) primarily in the initial hypothesis definition stage. The complete paper, code generated, specific agent logs, and the full checklist are available for review on OpenReview [5].

3 Observed Failure Modes and Mitigation

Through our experiments, we identified six critical failure modes that consistently emerged across different research attempts. These patterns reveal systematic limitations in current LLM capabilities for autonomous research. The following subsections detail each failure mode with specific examples and mitigation strategies.